REPUBLIC POLYTECHNIC

Lesson 10

Food Ordering Process Flow Simulation

E211 – Operations Planning II

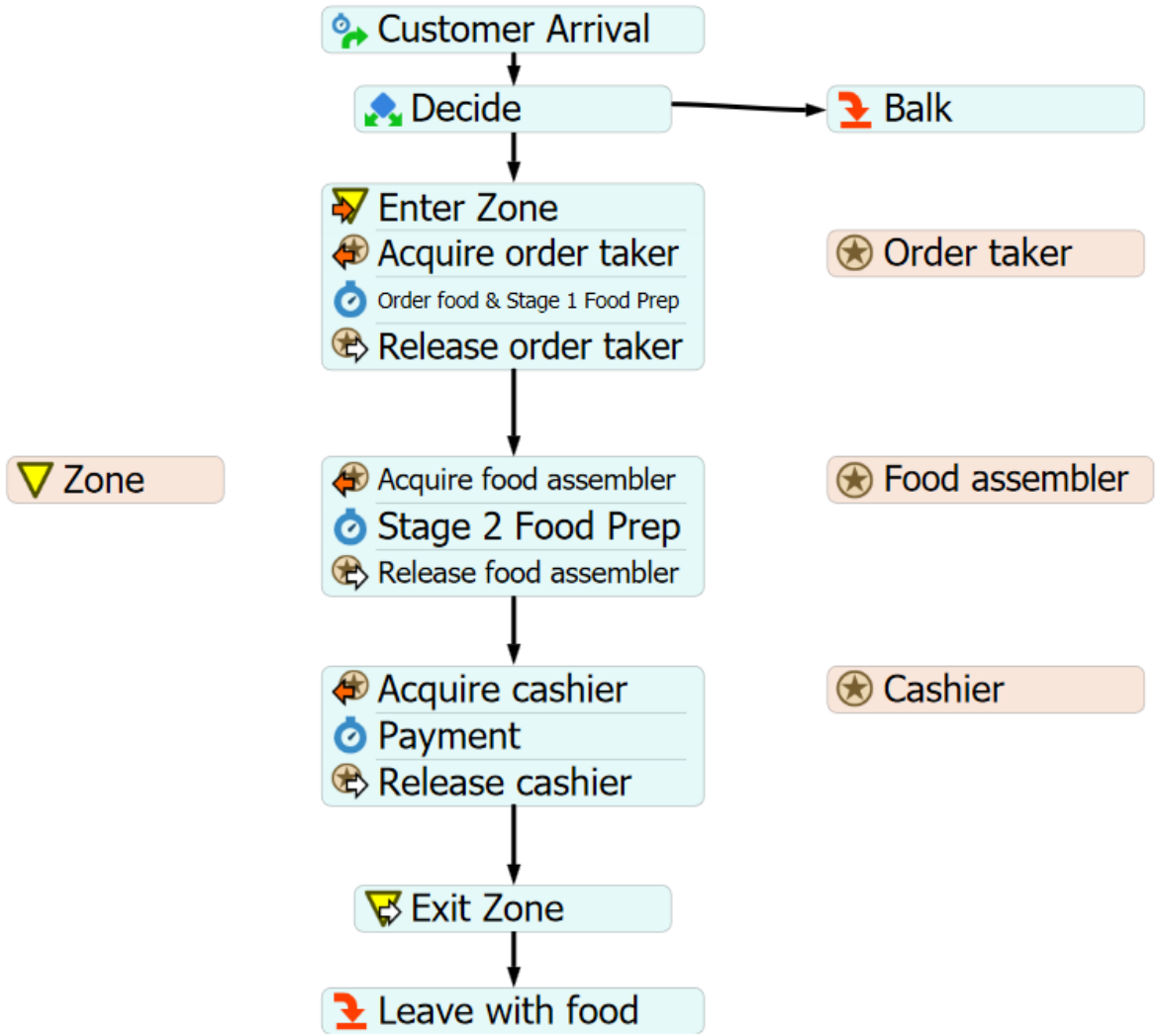
Learning Outcomes

At the end of the lesson, students should be able to:

- Construct a Process Flow model in FlexSim to simulate a multi-stage service system.
- Identify bottleneck in the process through simulation result analysis.
- Compare 2 sets of results using confidence interval of the difference between their mean performance measures

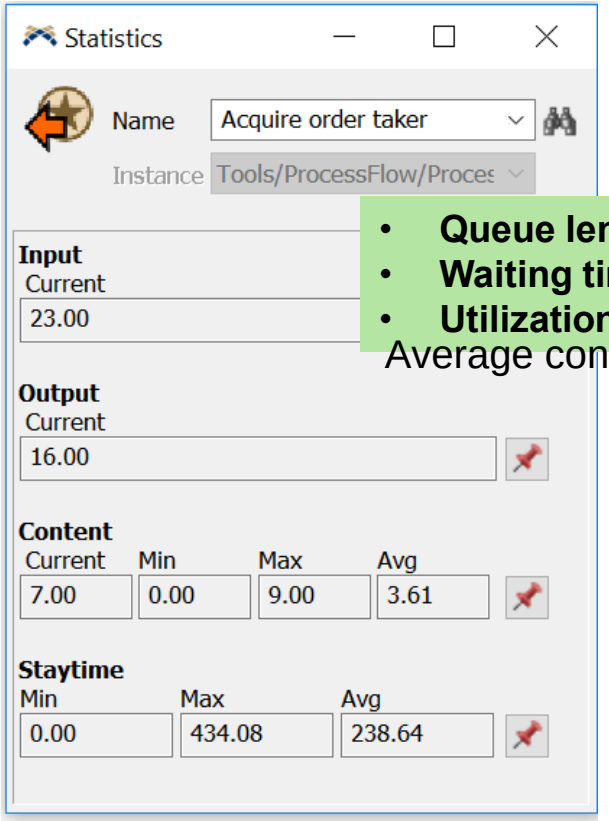
Food Ordering Process Flow

Process Flow Model in FlexSim



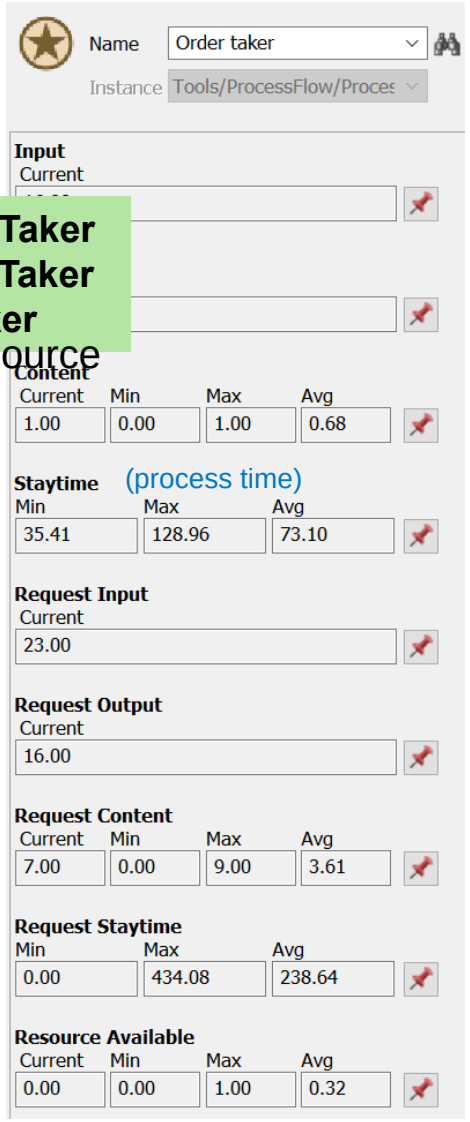
REPUBLIC POLYTECHNIC

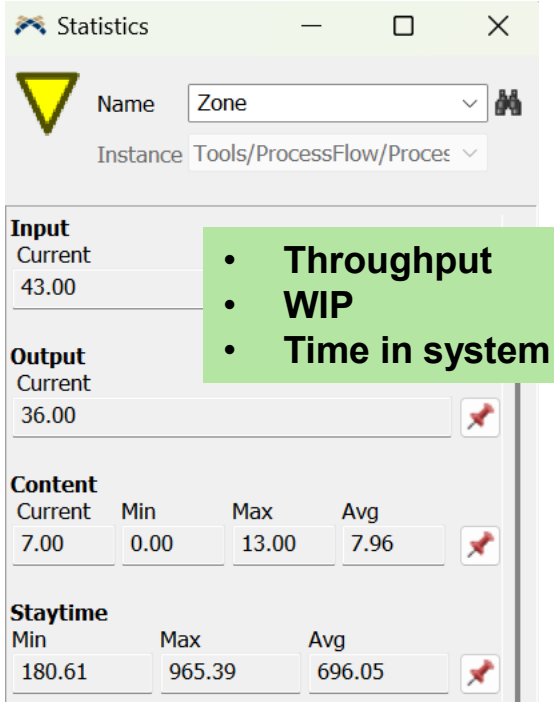
Know How to Interpret Activity Statistics



- **Queue length at Order Taker**
- **Waiting time for Order Taker**
- **Utilization of Order Taker**

Average content/Max Resource





- **Throughput**
- **WIP**
- **Time in system**

Queue length (OT): 7
Waiting Time (OT):
Utilization (OT):

request (part) - is the queue

1 - 0.68 = 0.32

Dashboard and Experimenter

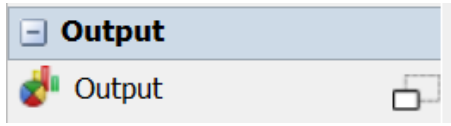
Performance Measures

Video resource:
https://tinyurl.com/msbexnfm

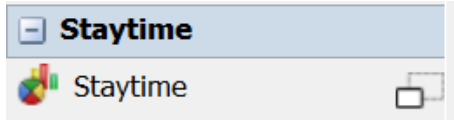
Create the following performance measures in a dashboard.

1. Number of Customers Served
2. Number of Customers Balked
3. Time in System
4. Waiting Time for Order Food
5. Waiting Time for Stage 2
6. Waiting Time for Payment

Use Output (table)



Use Staytime (table)



When adding a new chart, link to the activity in the Process Flow model using the ‘sampler’ function.

Remember to rename the chart title.

Performance Measures in Dashboard

ProcessFlow | Dashboard

Number of Customers Served

Object	Throughput
Leave with food	36

Waiting Time for Order Food

Object	Avg	Min	Max
Acquire order taker	6.70	0	12.09

Number of Customers Balked

Object	Throughput
Balk	2

Waiting Time for Stage 2

Object	Avg	Min	Max
Acquire food assembler	1.66	0	3.25

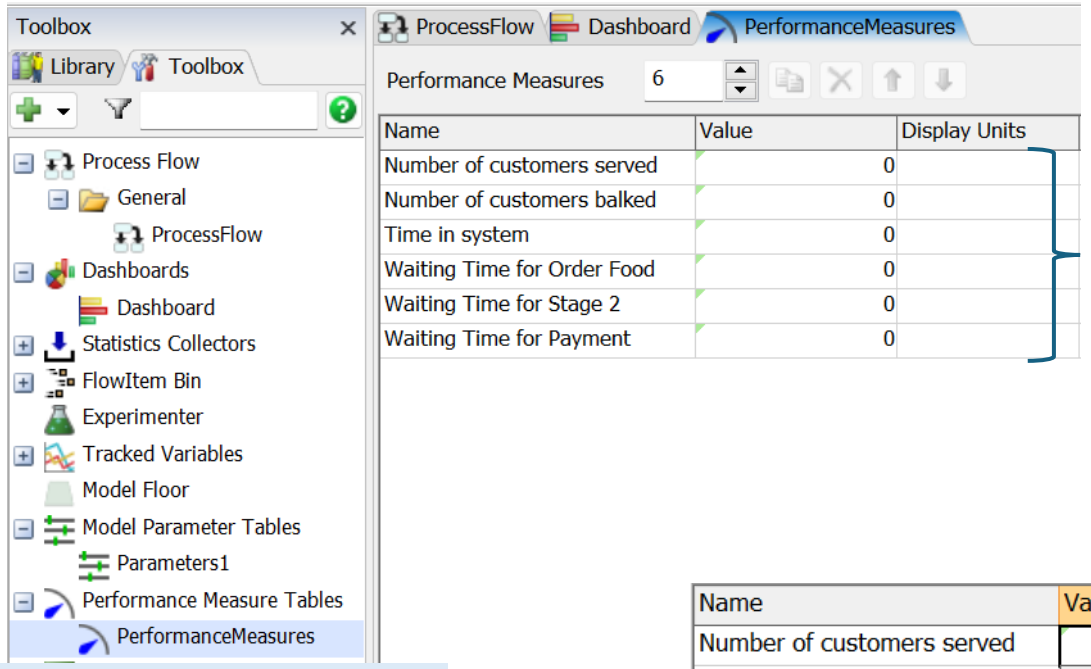
Time in System

Object	Avg	Min	Max
Zone	11.60	3.01	16.09

Waiting Time for Payment

Object	Avg	Min	Max
Acquire cashier	0	0	0

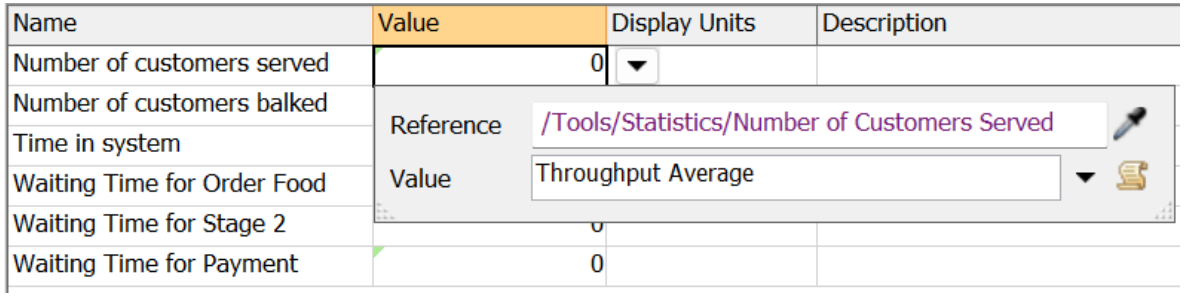
Create Performance Measures Table



Video resource:
https://tinyurl.com/2je7n83p

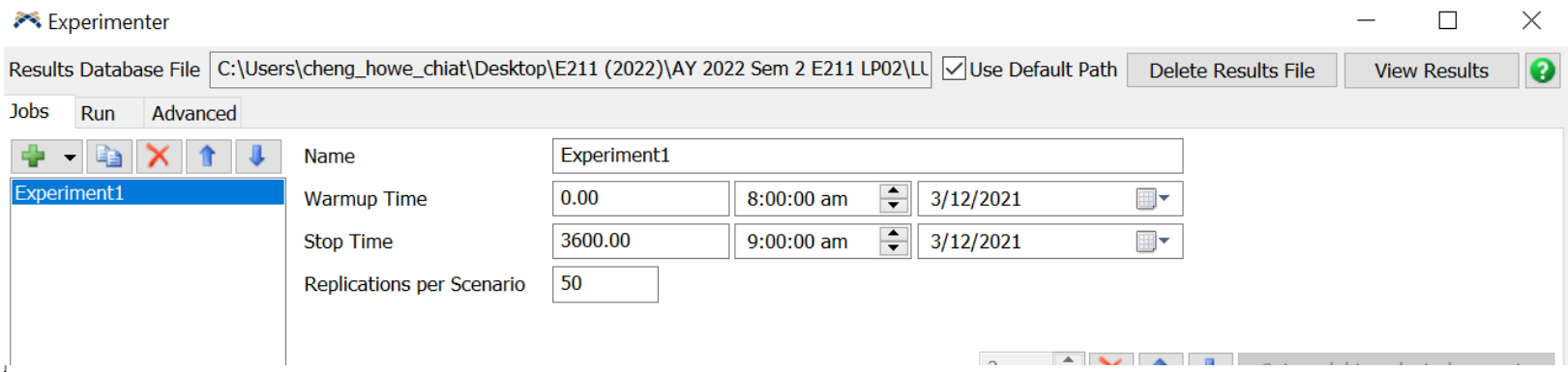
2. Create six Performance Measures and name them accordingly.

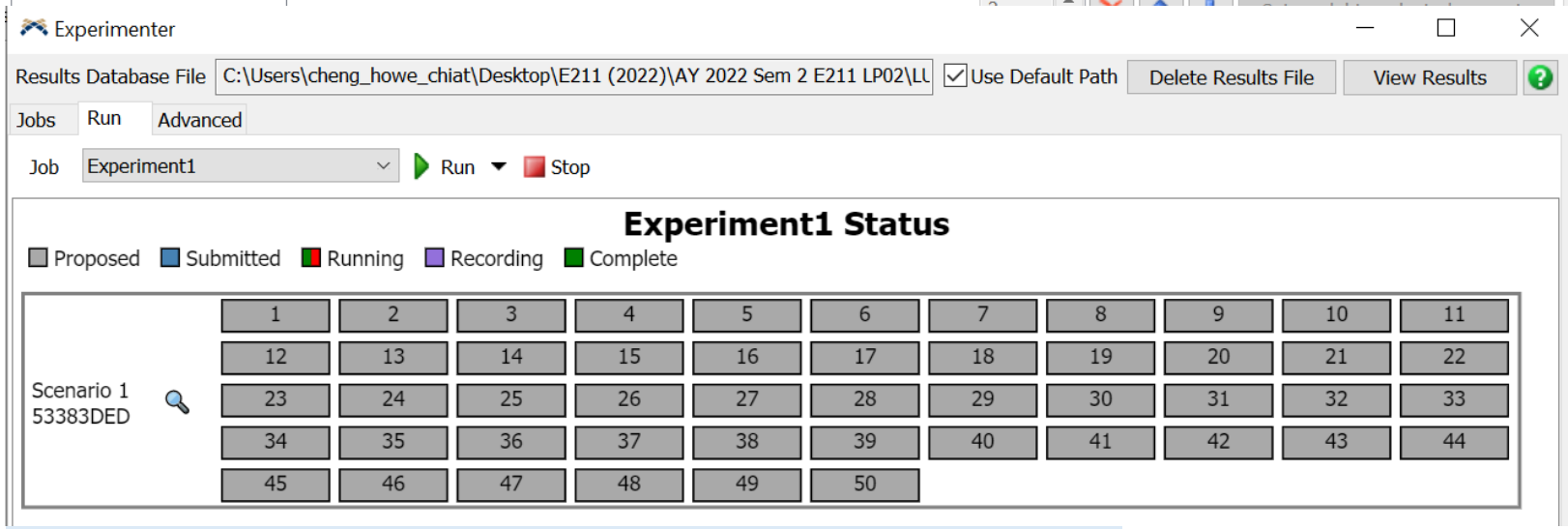
1. Go to Toolbox and open PerformanceMeasures Table.



3. For each Performance Measure, go to the Value cell and use Reference's Sampler tool to select the associated Dashboard chart (All Values to be Average).

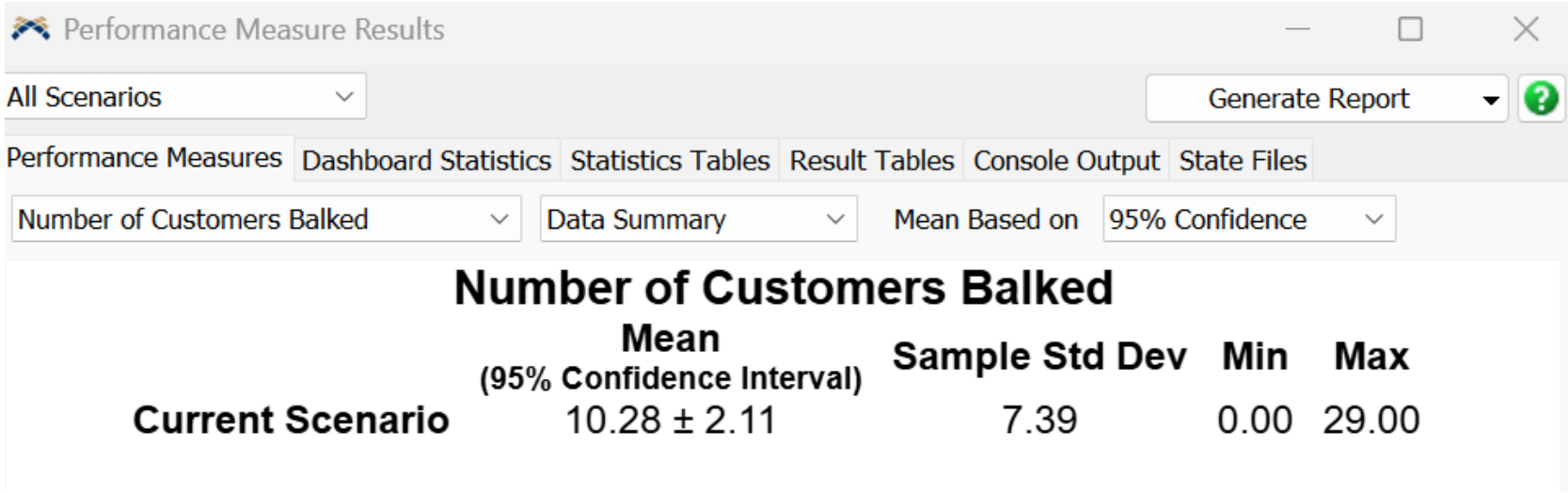
Experimenter Runs





1. Key in the Replications per Scenario (no. of simulation runs)
2. Key in the Stop Time
3. Go to Run tab and click Run
4. After Experimenter runs are completed, click View Results

Experimenter Result – Data Summary



Number of Customers Balked

	Mean (95% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	10.28 ± 2.11	7.39	0.00	29.00

- Data summary shows the mean, confidence interval of mean, standard deviation as well as min and max of the performance measure
- In the result shown, mean number of customers balked is 10.28, which is a lot higher than the result from a single simulation run!
- 95% confidence interval of Mean Number of Customers Balked is [8.17, 12.39]

10.28 – 2.11 10.28 + 2.11

Confidence Interval Results

Performance Measures | Dashboard Statistics | Statistics Tables | Result Tables | Console Output | State Files

Number of Customers Served | Data Summary | Mean Based on 95% Confidence

Number of Customers Served

	Mean (95% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	39.140 ± 0.473	1.654	35.000	42.000

Performance Measures | Dashboard Statistics | Statistics Tables | Result Tables | Console Output | State Files

Time in System | Data Summary | Mean Based on 95% Confidence

Time in System

	Mean (95% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	11.185 ± 0.364	1.274	7.429	13.337

Identify the Bottleneck

queue cannot be bottleneck

make sure queue

Compare Waiting Times	95% Confidence Interval of Mean (Minutes)
Waiting Time for Order Food	6.328 ± 0.394
Waiting Time for Food Assembly	1.669 ± 0.283
Waiting Time for Payment	0.00 ± N/A

Bottleneck is at Order Food since it has the highest average waiting time

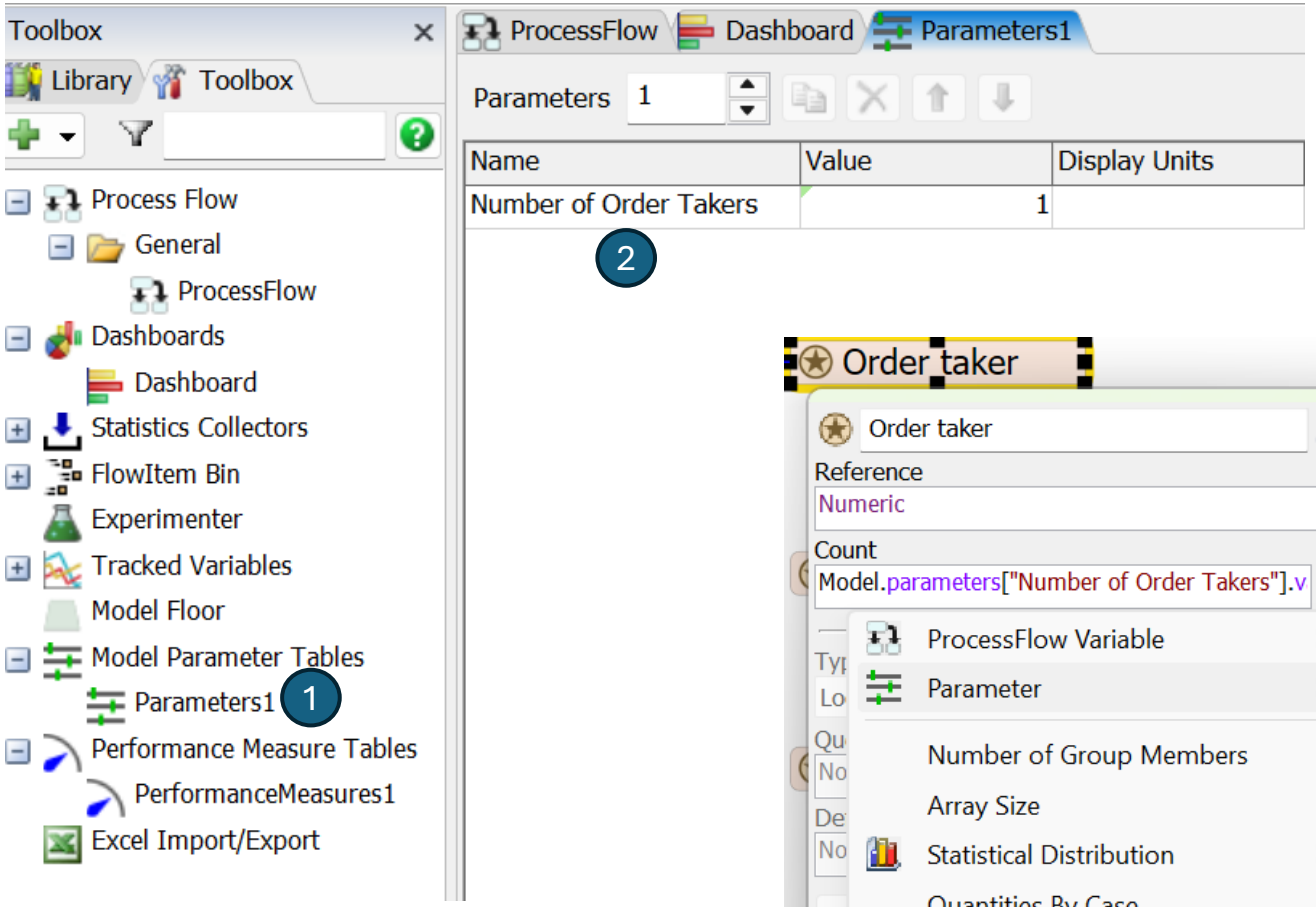
what is the solution?
add 1 more order taker
train the order taker (service times will be reduced)
digital order

Use of Parameters

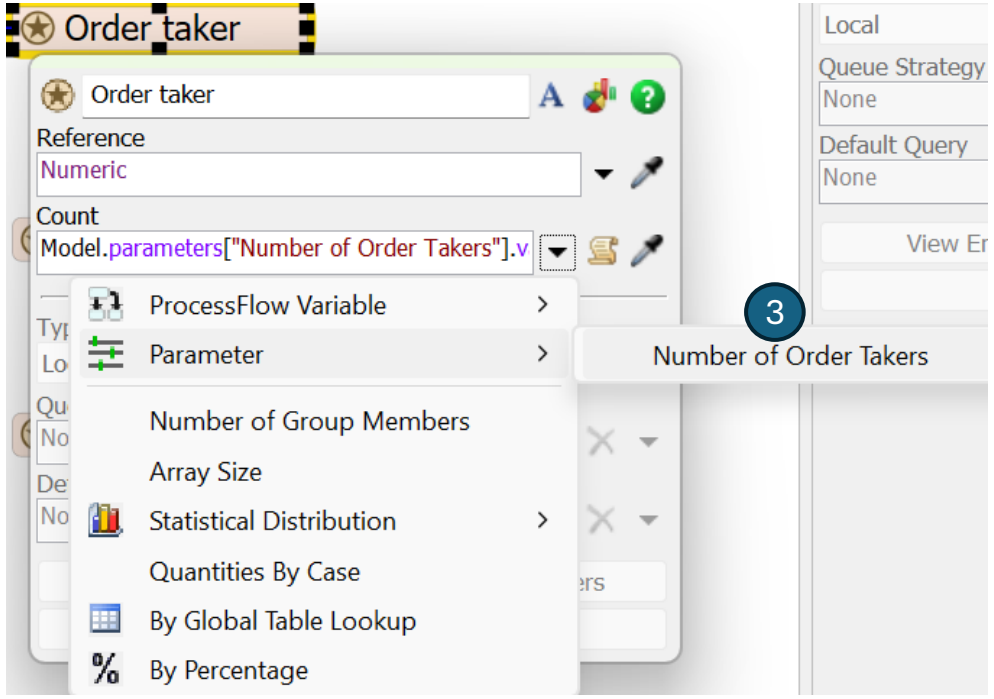
- What if there are 2 Order takers? How would it affect the performance measures?

- We can make use of Model Parameters to evaluate the impact of resource quantity on performance measures.

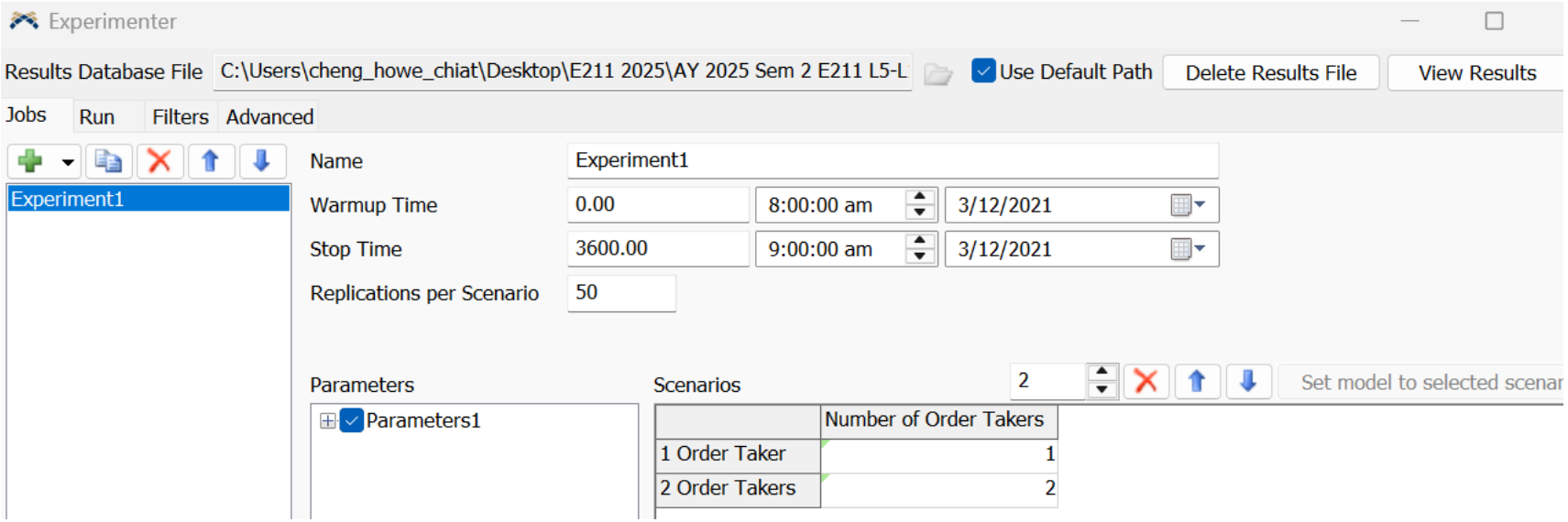
Set Up Model Parameters



1. Click on Parameters in Toolbox.
2. Rename the parameter.
3. Go to Order taker properties. Under Count, click Parameter and select 'Number of Order Takers'.

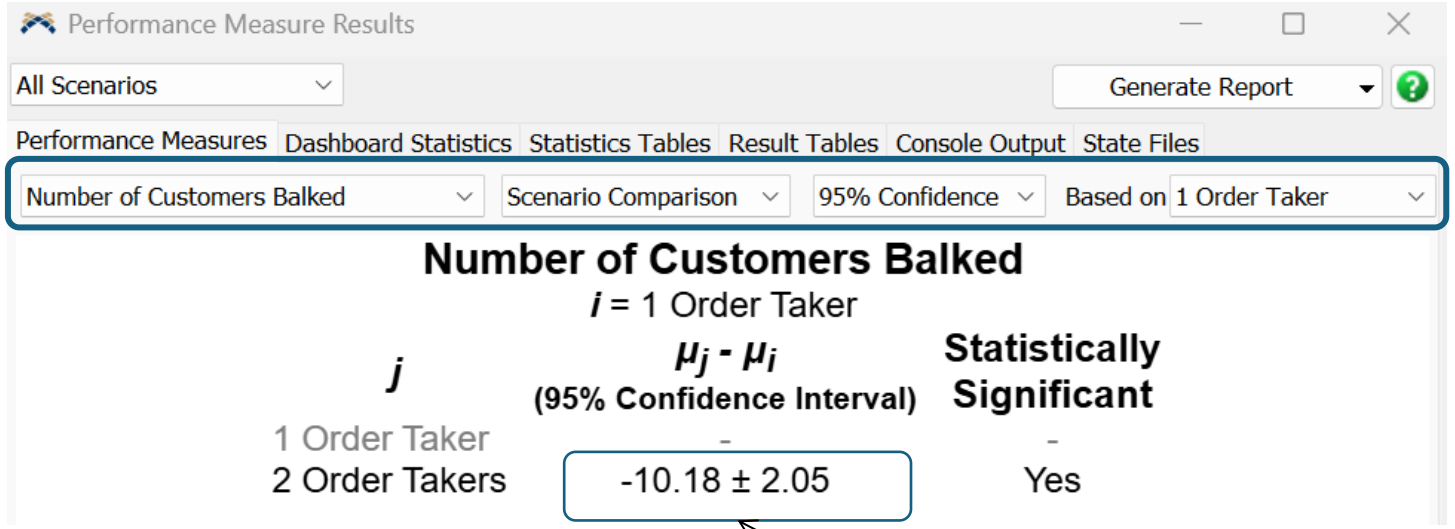


Set Up Scenarios in Experimenter



1. Go to Experimenter
2. Tick the Parameters box
3. Create 2 Scenarios as shown. Enter the correct Number of Order Takers for each scenario
4. Go to Run tab and run the Experimenter

Scenario Result Comparison



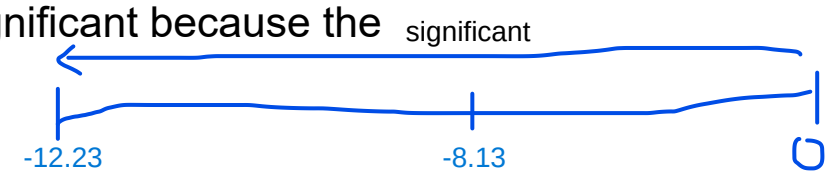
Number of Customers Balked

i = 1 Order Taker

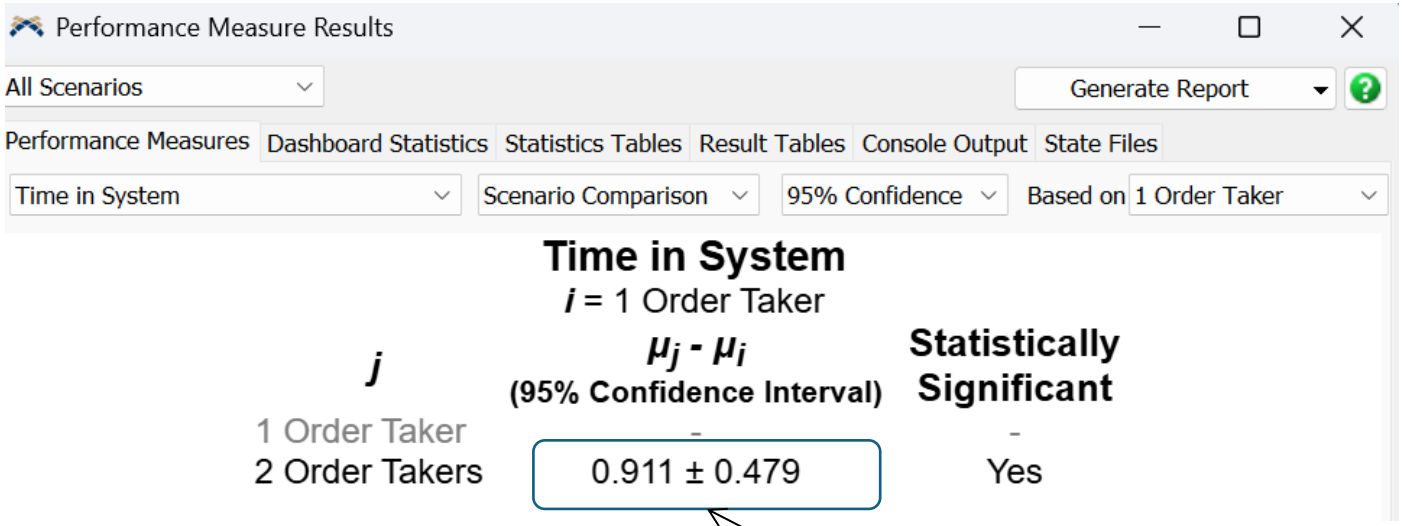
j	$\mu_j - \mu_i$ (95% Confidence Interval)	Statistically Significant
1 Order Taker	-	-
2 Order Takers	-10.18 ± 2.05	Yes

No. of Balks (2 Order Takers) – No. of Balks (1 Order Taker)

- Comparison of scenario (1 Order taker Vs 2 Order takers) shows that the difference in the mean Number of Customers Balked is statistically significant.
- The 95% confidence interval of the difference in Mean Number of Customers Balked is [-12.23, -8.13]. The difference is statistically significant because the confidence interval does not encompass the value of 0.



Scenario Result Comparison



Time in System
i = 1 Order Taker

j	$\mu_j - \mu_i$ (95% Confidence Interval)	Statistically Significant
1 Order Taker	-	-
2 Order Takers	0.911 ± 0.479	Yes

Time in System (2 Order Takers) – Time in System (1 Order Taker)

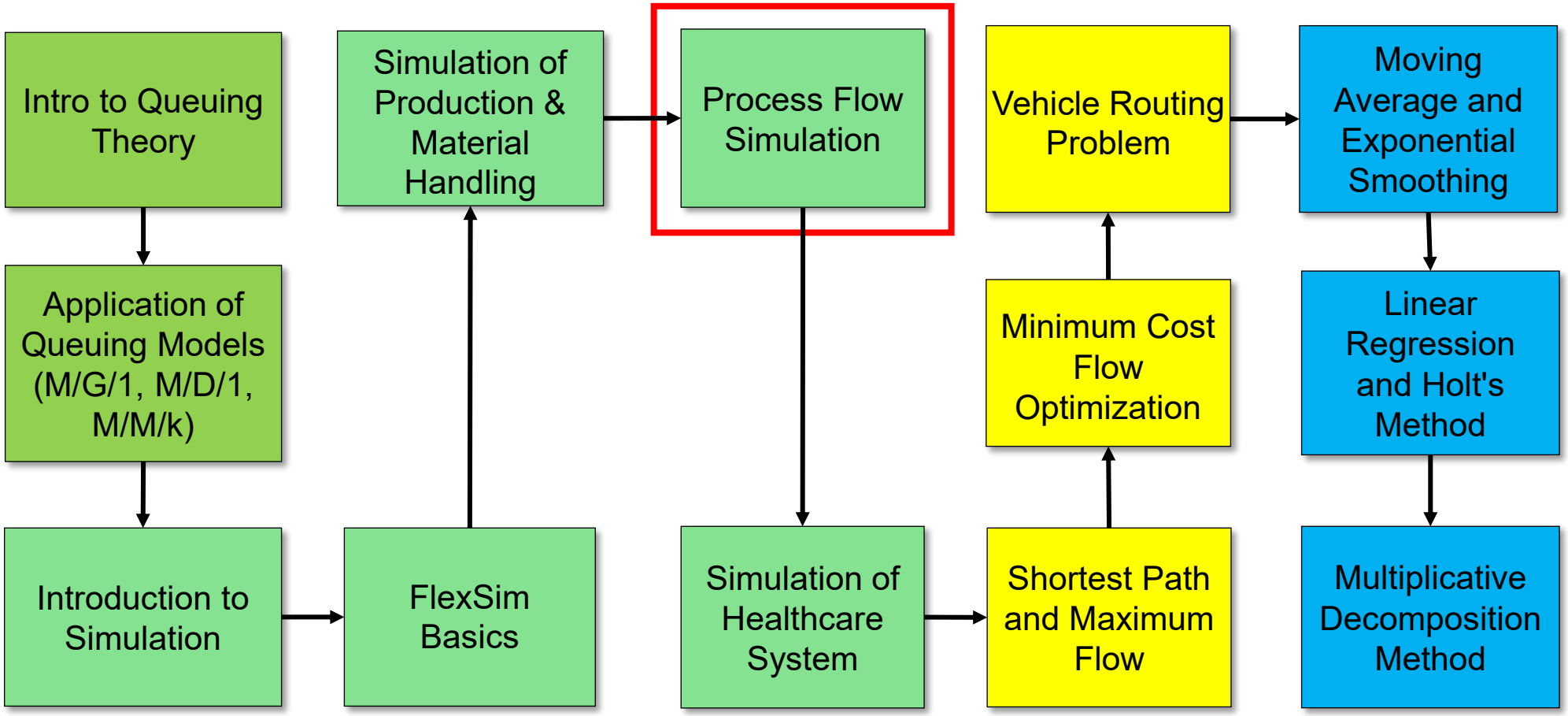
- Average time in system in the 2 Order Takers scenario is 0.911 minute higher than that in the 1 Order Taker scenario. Can you guess the reason why?

Experimentation

A. Determine the most efficient set up of staff and their role assignment to achieve these performance measures:

Mean Time in System ≤ 10mins
Mean Number of Customers served ≥ 45

B. Determine the total space required to accommodate customers in the system for your solution to part (A).

(We look at zone)

E211 Progression